

Module 3: Digital Transmission

PCM, DM, Line Encoding, Data rate



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M3L20: Digital Transmission

- Motivation
- Sampling
- Quantization
- PCM

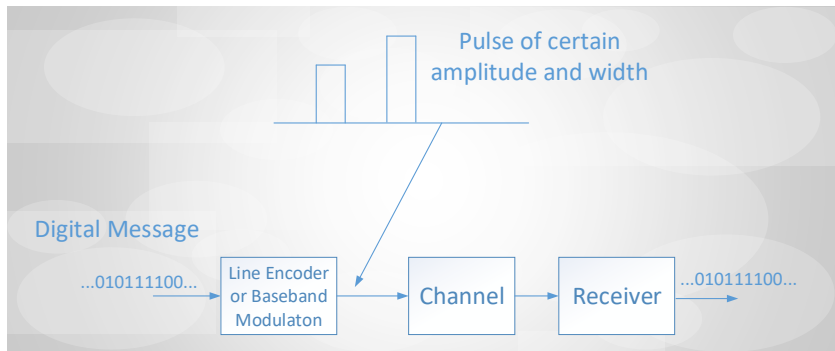


Information Source

- Human Speech
- Text Document
- Data

Digital Transmission: Motivation

- Line encoder converts the digital message into electrical pulses or waveforms for transmission over the channel.



American Standard Code for Information Interchange: ASCII

ASCII TABLE

Decimal	Hexadecimal	Binary	Octal	Char	Decimal	Hexadecimal	Binary	Octal	Char	Decimal	Hexadecimal	Binary	Octal	Char
0	0	0	0	[NULL]	48	30	110000	60	0	96	60	1100000	140	^
1	1	1	1	[START OF HEADING]	49	31	110001	61	1	97	61	1100001	141	a
2	2	10	2	[START OF TEXT]	50	32	110010	62	2	98	62	1100010	142	b
3	3	11	3	[END OF TEXT]	51	33	110011	63	3	99	63	1100011	143	c
4	4	100	4	[END OF TRANSMISSION]	52	34	110100	64	4	100	64	1100100	144	d
5	5	101	5	[ENQUIRY]	53	35	110101	65	5	101	65	1100101	145	e
6	6	110	6	[ACKNOWLEDGE]	54	36	110110	66	6	102	66	1100110	146	f
7	7	111	7	[BELL]	55	37	110111	67	7	103	67	1100111	147	g
8	8	1000	10	[BACKSPACE]	56	38	111000	70	8	104	68	1101000	150	h
9	9	1001	11	[HORIZONTAL TAB]	57	39	111001	71	9	105	69	1101001	151	i
10	A	1010	12	[LINE FEED]	58	3A	111010	72	:	106	6A	1101010	152	j
11	B	1011	13	[VERTICAL TAB]	59	3B	111011	73	:	107	6B	1101011	153	k
12	C	1100	14	[FORM FEED]	60	3C	111100	74	<	108	6C	1101100	154	l
13	D	1101	15	[CARRIAGE RETURN]	61	3D	111101	75	=	109	6D	1101101	155	m
14	E	1110	16	[SHIFT OUT]	62	3E	111110	76	>	110	6E	1101110	156	n
15	F	1111	17	[SHIFT IN]	63	3F	111111	77	?	111	6F	1101111	157	o
16		10000	20	[DATA LINK ESCAPE]	64	40	1000000	100	@	112	70	1110000	160	p
17	11	10001	21	[DEVICE CONTROL 1]	65	41	1000001	101	A	113	71	1110001	161	q
18	12	10010	22	[DEVICE CONTROL 2]	66	42	1000010	102	B	114	72	1110010	162	r
19	13	10011	23	[DEVICE CONTROL 3]	67	43	1000011	103	C	115	73	1110011	163	s
20	14	10100	24	[DEVICE CONTROL 4]	68	44	1000100	104	D	116	74	1110100	164	t
21	15	10101	25	[NEGATIVE ACKNOWLEDGE]	69	45	1000101	105	E	117	75	1110101	165	u
22	16	10110	26	[SYNCHRONOUS IDLE]	70	46	1000110	106	F	118	76	1110110	166	v
23	17	10111	27	[END OF TRANS. BLOCK]	71	47	1000111	107	G	119	77	1110111	167	w
24	18	11000	30	[CANCEL]	72	48	1001000	110	H	120	78	1111000	170	x
25	19	11001	31	[END OF MEDIUM]	73	49	1001001	111	I	121	79	1111001	171	y
26	1A	11010	32	[SUBSTITUTE]	74	4A	1001010	112	J	122	7A	1111010	172	z
27	1B	11011	33	[ESCAPE]	75	4B	1001011	113	K	123	7B	1111011	173	{
28	1C	11100	34	[FILE SEPARATOR]	76	4C	1001100	114	L	124	7C	1111100	174	
29	1D	11101	35	[GROUP SEPARATOR]	77	4D	1001101	115	M	125	7D	1111101	175	}
30	1E	11110	36	[RECORD SEPARATOR]	78	4E	1001110	116	N	126	7E	1111110	176	~
31	1F	11111	37	[UNIT SEPARATOR]	79	4F	1001111	117	O	127	7F	1111111	177	[DEL]
32	20	100000	40	[SPACE]	80	50	1010000	120	P					
33	21	100001	41	!	81	51	1010001	121	Q					
34	22	100010	42	"	82	52	1010010	122	R					
35	23	100011	43	#	83	53	1010011	123	S					
36	24	100100	44	\$	84	54	1010100	124	T					
37	25	100101	45	%	85	55	1010101	125	U					
38	26	100110	46	&	86	56	1010110	126	V					
39	27	100111	47	'	87	57	1010111	127	W					
40	28	101000	50	(	88	58	1011000	130	X					
41	29	101001	51	)	89	59	1011001	131	Y					
42	2A	101010	52	*	90	5A	1011010	132	Z					
43	2B	101011	53	+	91	5B	1011011	133	[					
44	2C	101100	54	,	92	5C	1011100	134	\					
45	2D	101101	55	-	93	5D	1011101	135	]					
46	2E	101110	56	.	94	5E	1011110	136	^					
47	2F	101111	57	/	95	5F	1011111	137	_					

Digit to Waveform

Digit	Binary equivalent	Pulse code waveform
0	0000	
1	0001	
2	0010	
3	0011	
4	0100	
5	0101	
6	0110	
7	0111	
8	1000	
9	1001	
10	1010	
11	1011	
12	1100	
13	1101	
14	1110	
15	1111	

M3L21: Digital Transmission

- Nyquist Criteria
- PCM
- DM



Channel Capacity: Nyquist Criteria for Data Transmission

- Scenario: Channel with BW B_{channel} and noise-free
- Map binary digits to waveform/pulse/symbol/signal
- Maximum channel capacity or rate of information transfer:
 $R_s = 2B_{\text{channel}}$ symbols per second.
- Note: This Nyquist criteria is different from Nyquist sampling theorem

Channel Bandwidth Requirement: Analog versus Digital

if message signal has a bandwidth B

Analog Transmission requires channel bandwidth as follows:

- AM $B_{\text{channel}} = 2B$
- SSB $B_{\text{channel}} = B$
- FM $B_{\text{channel}} = 2(\Delta_f + B)$

What will be channel bandwidth requirement for digital transmission?

Given B_{channel} , how many waveform you can transmit in a second:
 $R_s = 2B_{\text{channel}}$ symbols per second.

Data Transmission Rate

- If waveform/pulse/symbol/ baud rate is $R_s = 2B_{\text{channel}}$; what is data rate transmission in bps?
- How many bits a waveform carry?
- For one bit per waveform transmission: $M = 2$ (i.e. binary) different wave forms are required: $R_b = R_s$ bps
- Bit patterns: $1 \rightarrow p_1(t)$, $0 \rightarrow p_2(t)$
- For two bits per waveform transmission: $M = 4$ (i.e. 4-ary) different wave forms are required: $R_b = 2R_s$ bps
- For n bits per waveform transmission: $M = 2^n$ (i.e. M -ary) different wave forms are required: $R_b = nR_s$ bps
- Bit patterns: $111111... \rightarrow p_1(t), ... 00000000.. \rightarrow p_M(t)$

R_b versus R_s

- $M = 2^n$
- $n = \log_2 M$
- $R_b = nR_s$

What to study in digital transmission?

- Is there any limit on M ? of course $M = 2$ is the minimum.
- How to choose $p_1(t)$, $p_2(t)$, ..., $p_M(t)$
- Most important, how to design receiver to decode binary bit stream

CW Problem

A channel bandwidth of 100 Hz is available. What is waveform rate/pulse rate/ baud rate/ signalling rate/symbol rate

Answer: $R_s = 200$ symbols per second.

What is data rate in bits per second?

Answer: $R_b = 200$ bps

CW Problem

How much time it will take to transmit a 10-page report over a channel having bandwidth (a) 10KHz (b) 20MHz. Assume binary and 16-ary transmission.

Solution (a)

10-page contains approximately $10 \times 3000 = 30000$ characters
ASCII Coding: $3000 \times 7 = 210000$ bits.

- $R_s = 20,000$ symbols per second.
- For binary transmission: $R_b = 20,000$ bps.
- Time taken $210000/20000 = 10.5$ seconds.
- For 16-ary transmission: $R_b = 4R_s = 80,000$ bps
- Time Taken $210000/80000 = 2.625$ seconds.

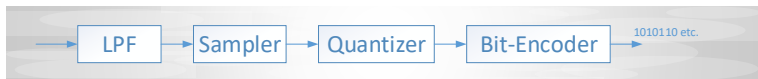
Solution (b)

10-page contains approximately $10 \times 3000 = 30000$ characters
ASCII Coding: $3000 \times 7 = 210000$ bits.

- $R_s = 40 \times 10^6$ symbols per second.
- For binary transmission: $R_b = 40 \times 10^6$ bps.
- Time taken $210000/(40 \times 10^6) = 5.25$ milli second.
- For 16-ary transmission: $R_b = 4R_s = 160 \times 10^6$ bps
- Time Taken $210000/(160 \times 10^6) = 1.3125$ milli second.

Pulse Code Modulation (PCM)

- PCM: Convert analog signal into digital for transmission.
- PCM modifies the modulating signal without carrier.
- PCM: A/D



- PCM: Baseband transmission



PCM: Sampling and Quantization

Take a message signal $m(t)$, say it has maximum 8 minimum -8

- Divide -8 to 8 in $L = 2^N$ parts
- for $N = 1, L = 2$;

Interval	Midpoint	Binary Code
$[-8, 0)$	-4	0
$[0, 8]$	$+4$	1

- Mid-point is the quantized value

Note: Binay code is not decimal equivalent of mid-point

PCM: Sampling and Quantization

Take a message signal $m(t)$, say it has maximum 8 minimum -8

- for $N = 2, L = 4$;

Interval	Midpoint	Binary Code
$[-8, -4)$	-6	00
$[-4, 0)$	-2	01
$[0, 4)$	$+2$	10
$[4, 8]$	$+6$	11

Mid-point is the quantized value

Note: Binay code is not decimal equivalent of mid-point

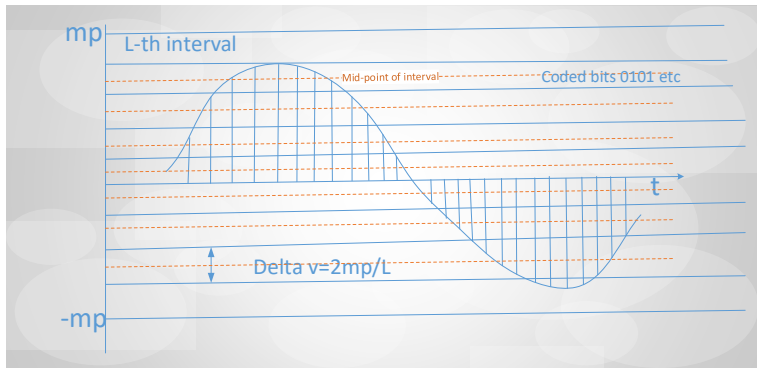
PCM: Sampling and Quantization

Take a message signal $m(t)$, say it has maximum 8 minimum -8

- for $N = 3, L = 8$;

Interval	Midpoint	Binary Code
$[-8, -6)$	-7	000
$[-6, -4)$	-5	001
$[-4, -2)$	-3	010
$[-2, 0)$	-1	011
$[0, 2)$	$+1$	100
$[2, 4)$	$+3$	101
$[4, 6)$	$+5$	110
$[6, 8]$	$+7$	111

PCM: Sampling and Quantization



Data rate generation examples

What is source data rate generation of a sine signal of 10 Hz?

What is source data rate generation of a voice signal?

What is source data rate generation of a video signal?

M3L22: Digital Transmission

- PCM
- DM
- Nyquist Pulse



PCM: Sampling and Quantization

- Quantization error/noise: $q(t) = m(t) - \hat{m}(t)$.
- We receive $y(t) = m(t) + q(t)$, where $q(t)$: quantization noise.

PCM: Quantization Noise Power

- Maximum quantization error $\pm \frac{\Delta v}{2}$: since we take mid-point of levels $\Delta v = \frac{2m_p}{L}$.
- Quantization error lies in the range $(-\Delta v/2, \Delta v/2)$
- Model quantization error/noise as $q \sim U(-\Delta v/2, \Delta v/2)$
- PDF of uniform $f_Q(q) = \frac{1}{b-a} = \frac{1}{\Delta v}$
- Variance of uniform RV will be power of quantization noise:
- $\overline{q^2} = \frac{1}{\Delta v} \int_{-\Delta v/2}^{\Delta v/2} q^2 dq$
- $\overline{q^2} = \frac{(\Delta v)^2}{12}$
- $\overline{q^2} = \frac{m_p^2}{3L^2}$
- $N_q = \overline{q^2(t)} = \overline{q^2} = \frac{m_p^2}{3L^2}$

PCM: SQNR

- $y(t) = m(t) + q(t)$
- Signal power $S_0 = P_m$
- Noise power $N_0 = N_q = \overline{q^2(t)} = \overline{q^2} = \frac{m_p^2}{3L^2}$
- SQNR $\frac{S_0}{N_0} = 3L^2 \frac{P_m}{m_p^2}$

PCM: 6-dB rule

- $\text{SQNR} = 3L^2 \frac{P_m}{m_p^2}$
- $L = 2^N$
- $\text{SQNR} = 3 \times 2^{2N} \frac{P_m}{m_p^2}$
- $\text{SQNR in dB} = 10 \log_{10} 3 \frac{P_m}{m_p^2} + 2N \times 10 \log_{10} 2$
- $10 \log_{10} 2 = 3.010 = 3$
- $\text{SQNR in dB} = 10 \log_{10} 3 \frac{P_m}{m_p^2} + 6N$
- One bit increase increases the SQNR by 6-dB: 6dB Rule

Data Rate and BW of for PCM

- $R_b = 2Nf_m \text{ bits/sec}$
- Channel BW for PCM transmission (binary):
 $B = R_b/2 = Nf_m \text{ Hz}$
- Channel BW for PCM transmission ($M = 4$): ?

CW Problem

A message signal $m(t)$ is transmitted by binary PCM without compression. If the SQNR is required to be at least 47 dB, determine the minimum value of $L = 2^N$ required, assuming that $m(t)$ is sinusoidal. Determine the actual SQNR obtained with this minimum L .

Solution

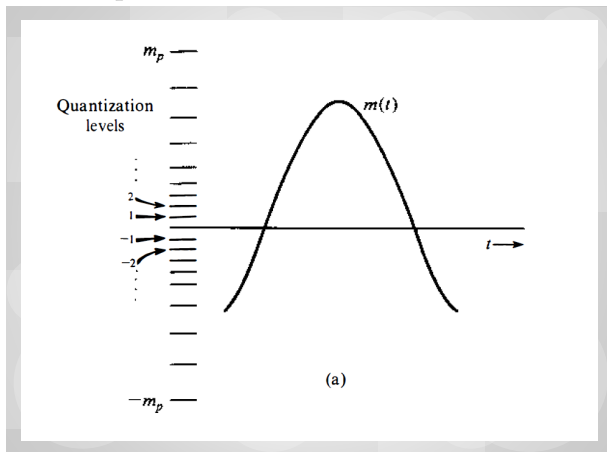
- $\frac{P_m}{m_p^2} = 0.5$
- SQNR 47dB = 50119
- $3L^2 \frac{\overline{m^2(t)}}{m_p^2} \geq 50119$
- $L \geq 182.8$
- $L = 256 = 2^8$
- Actual SQNR $3L^2 \frac{\overline{m^2(t)}}{m_p^2} = 98304 = 49.43dB$

- 6.2-1** The American Standard Code for Information Interchange (ASCII) has 128 characters, which are binary-coded. If a certain computer generates 100,000 characters per second, determine the following:
- (a) The number of bits (binary digits) required per character.
 - (b) The number of bits per second required to transmit the computer output, and the minimum bandwidth required to transmit this signal.

- 6.2-2** A compact disc (CD) records audio signals digitally by using PCM. Assume that the audio signal bandwidth equals 15 kHz.
- (a) If the Nyquist samples are uniformly quantized into $L = 65,536$ levels and then binary-coded, determine the number of binary digits required to encode a sample.
 - (b) If the audio signal has average power of 0.1 watt and peak voltage of 1 volt. Find the resulting signal-to-quantization-noise ratio (SQNR) of the uniform quantizer output in part (a).
 - (c) Determine the number of binary digits per second (bit/s) required to encode the audio signal.
 - (d) For practical reasons discussed in the text, signals are sampled at a rate well above the Nyquist rate. Practical CDs use 44,100 samples per second. If $L = 65,536$, determine the number of bits per second required to encode the signal, and the minimum bandwidth required to transmit the encoded signal.

PCM: Non-Uniform Quantization

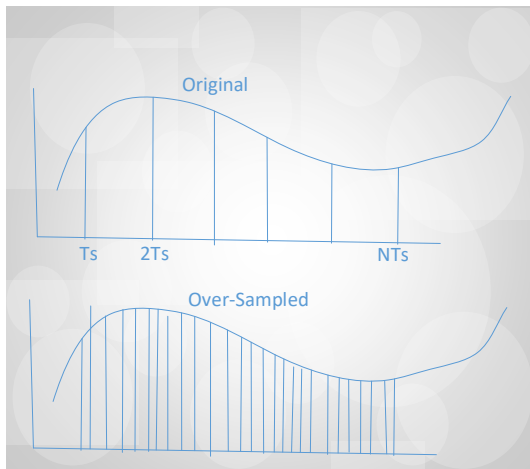
- SQNR $\frac{S_0}{N_0} = 3L^2 \frac{\overline{m^2(t)}}{m_p^2}$
- For better performance, SQNR needs to be constant.
- Do non-uniform quantization:



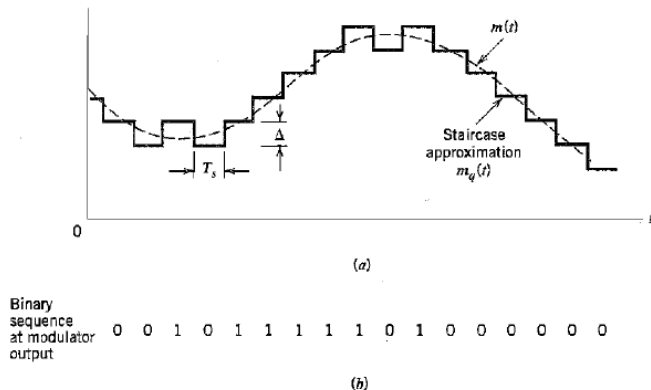
Delta Modulation

- 2 level of encoding: Delta Modulation.

Delta Modulation: Oversampling

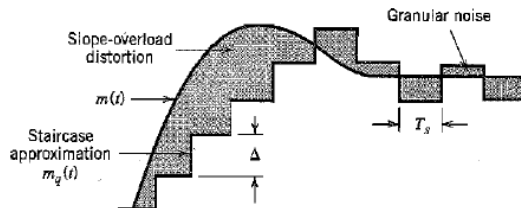


Delta Modulation: How it works



Notation: Use t_s

Delta Modulation Noise: Slope Overload and Granular noise



Notation: Use t_s

Delta Modulation: Slope Overload

- Slope over load occurs if $m_q(t)$ does not follow $m(t)$.
- No Slope overload condition: $\frac{\Delta}{t_s} \geq \max \frac{dm(t)}{dt}$.

Delta Modulation: Granular noise

- If step size is too large relative to the local slope of $m(t)$
- It is similar to the quantization.
- Granular noise is distributed between $-\Delta$ to Δ
- Range $(-\Delta, \Delta)$: uniformly distributed.
- Granular noise power: $\frac{\Delta^2}{3}$.

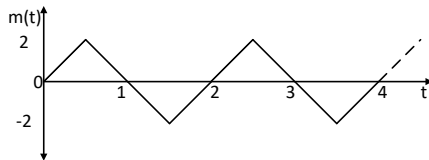
Slope Overload: Example

- Find the no over load condition for $m(t) = A \cos(2\pi f_m t)$

Adaptive Delta Modulation (ADM) dynamically varies the step size Δ according to the rate of change of the input signal, using larger Δ for steep slopes (to avoid slope overload) and smaller Δ for flat regions (to reduce granular noise).

CW Problem:

A periodic message signal $m(t)$ (as shown in figure) is to be converted to binary bit-stream using PCM and DM techniques.



- 1 The signal is sampled at a rate of 1000 samples per second and encoded using the PCM. Find the minimum bandwidth required to transmit the data to achieve an SQNR of 256.
- 2 Find the minimum step-size of the DM to avoid slope-over load condition achieving the same data rate transmission as that of the PCM (as obtained in part (a)).

Solution

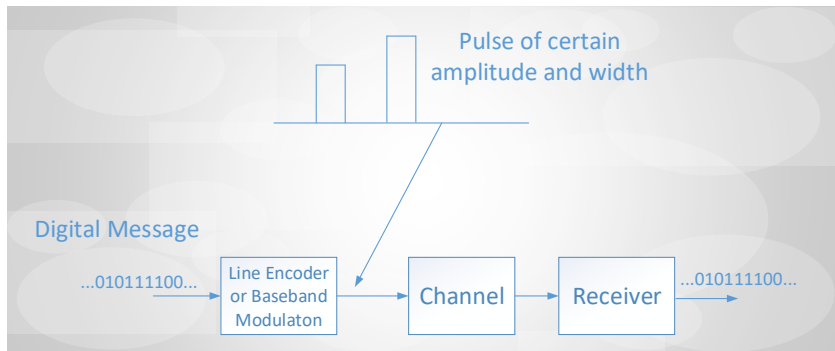
Average Power = $\frac{4}{3}$, $m_p = 2$, $SQNR = 3L^2 \frac{P_m}{m_p^2}$, which gives $L = 16$
= 4 bits/sample. Total data rate = 4000 bits/sec. BW=2000 Hz.

Solution

$$f_s = 4000 \text{ samples/sec. } \frac{dm(t)}{dt} = 4, \Delta \geq \frac{1}{1000}.$$

Baseband Digital Transmission: Line Encoding

- Line encoder converts the digital message into electrical pulses or waveforms for transmission over the channel.



Types of Line Encoding

- Return to Zero (RZ): Pulses return to zero before the end of the bit.
- RZ pulses have width less than T_b , mostly half-width.
- Non-return to Zero (NRZ): Pulses do not return to zero before the end of the bit.
- NRZ pulses have full width.

Types of Binary Line Encoding

- On-Off: $1 \rightarrow p(t)$, $0 \rightarrow 0$
- Polar: $1 \rightarrow p(t)$, $0 \rightarrow -p(t)$
- Bipolar or pseudo-ternary or alternate mark inversion (AMI):
0 is encoded as no pulse, $1 \rightarrow p(t)$ if previous bit 1 was coded to $-p(t)$ or $1 \rightarrow -p(t)$ if previous bit 1 was coded to $p(t)$.
Thus 1's alternate in polarity.
- Split-phase or Manchester: 1 to $p(t)$ for first half-width and $-p(t)$ for next half-width, 0 to $-p(t)$ for first half-width and $p(t)$ for next half-width.

Line Encoding: Example

Figure 7.2

Line code

examples:

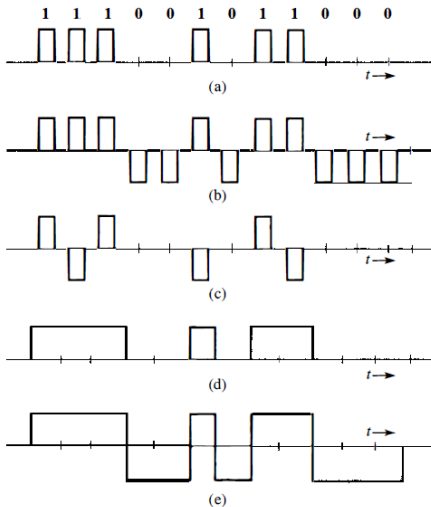
(a) on-off (RZ);

(b) polar (RZ);

(c) bipolar (RZ);

(d) on-off (NRZ);

(e) polar (NRZ).



Band limited Channels

- What will be the effect if an ideal rectangular pulse (time-limited) is transmitted over a band-limited channel?
- What will be the effect if a BW-limited (time-unlimited) is transmitted over a band-limited channel?
- ISI

Criterion for Zero Intersymbol Interference

- Simple technique: Transmits pulses that are not overlapping
- Is it possible to have non-overlapping pulses?
- Yes or No
- Nyquist criteria: Allowed to pulses to overlap but without ISI
- Nyquist pulse for zero ISI:

$$p(t) = \begin{cases} 1, t = 0 \\ 0, t = \pm nT_s, n \neq 0 \end{cases}$$

- Can you suggest such a pulse?

Nyquist Criterion for Zero Intersymbol Interference

- Nyquist criterion for zero ISI (Time-Domain):

$$p(nT_s) = \begin{cases} 1, n = 0 \\ 0, n \neq 0 \end{cases}$$

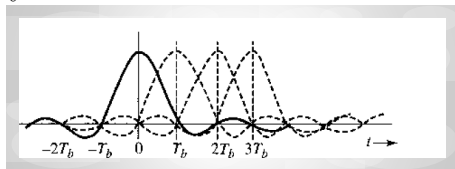
- Nyquist criterion for zero ISI (Frequency-Domain):
- $\sum_{n=-\infty}^{\infty} P(f + \frac{n}{T}) = T$
- Sum of spectra formed by repeating $P(f)$ spaced $1/T_s$ or R_s apart must be a constant.

Criterion for Zero Intersymbol Interference: sinc pulse

- There are many possibilities for $p(t)$.
- However, we need to limit the BW of that pulse to $R_s/2$
- A pulse which satisfies the Nyquist criterion for zero ISI and has BW $R_s/2$:

$$p(t) = \text{sinc}(\pi R_s t) = \begin{cases} 1, & t = 0 \\ 0, & t = \pm nT_s \end{cases}$$

- This is the only one choice.



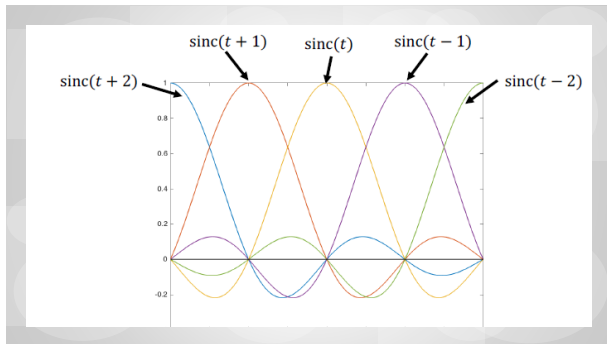
Read T_s

Problems with sinc pulse

- Problem 1: sinc pulse starts at $-\infty$: need to wait an infinite time to generate it.
- Solution to problem 1: Truncate the time samples
- Problem to solution to problem 1: increases BW beyond $R_s/2$.
- Problem 2: It decays too slowly at a rate $1/t$.
- In non-ideal scenario (practical scenario), a little deviation from the sampling instant will cause ISI.
- The problem 2 is the major concern in a practical setup.

Zero-ISI pulse with minimum BW: Sinc Pulse

- Problem : It decays too slowly at a rate $1/t$.



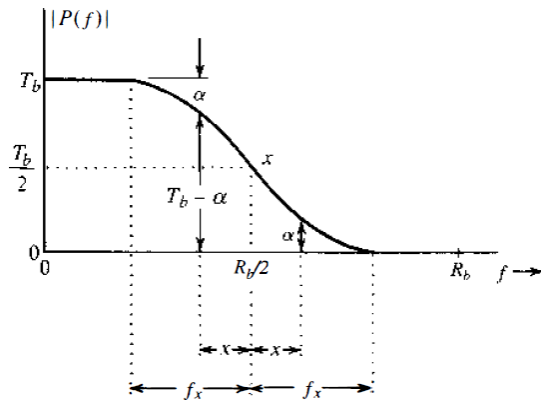
Nyquist Pulse with Zero ISI and Faster Decaying Rate

- Nyquist solved the decaying problem of sinc pulse.
- Raised cosine pulse (where α is the roll-off factor):
- $p(t) = \frac{\cos(\pi\alpha R_s t)}{1-(2\alpha R_s t)^2} \text{sinc}(\pi R_s t)$

$$P(f) = \begin{cases} T, & |f| \leq \frac{1-\alpha}{2T} \\ \frac{T}{2} \left(1 + \cos\left(\frac{\pi T}{\alpha} \left(|f| - \frac{1-\alpha}{2T} \right) \right) \right), & \frac{1-\alpha}{2T} < |f| < \frac{1+\alpha}{2T} \\ 0, & \text{elsewhere} \end{cases}$$

- where $T = 1/R_s$.
- Can you show that pulse $p(t)$ decays at a rate of $1/t^3$.
- $B_T = \frac{1+\alpha}{2T_s}$
- Higher α , more BW but faster decaying rate.

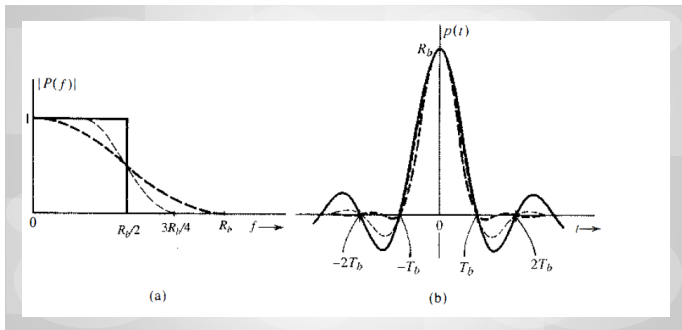
Excess BW



- $B_T = \frac{1+\alpha}{2T_s}$

Nyquist Pulse with Zero ISI

- Plot $P(f)$ for $\alpha = 0, \alpha = 0.5, \alpha = 1$.
- What is B_T for $\alpha = 0, \alpha = 0.5, \alpha = 1$.
- Plot $p(t)$ for $\alpha = 0, \alpha = 0.5, \alpha = 1$.



CW Problem

A computer puts out binary data at the rate of 56 kpps. The computer output is transmitted using a baseband binary PAM system that is designed to have a raised-cosine spectrum. Determine the transmission bandwidth and excess bandwidth for roll-off factors $\alpha = 0.25, 0.5, 0.75, 1$.

Solution 1

- $R_b = 56 \text{ Kpps}$
- What is minimum bandwidth?
- $B_T(\min) = R_b/2 = 28 \text{ KHz}$.
- Which pulse?
- Sinc pulse
- Frequency spectrum?
- Ideal rectangular
- Why to proceed further if sinc pulse does the job?
- Longer time to generate sinc pulse, and decaying of sinc pulse is $1/t$.
- Raised cosine pulse
- $p(t) = \frac{\cos(\pi\alpha R_b t)}{1-(2\alpha R_b t)^2} \text{sinc}(\pi R_b t)$

$$P(f) = \begin{cases} T, & |f| \leq \frac{1-\alpha}{2T} \\ \frac{T}{2} \left(1 + \cos\left(\frac{\pi T}{\alpha} \left(|f| - \frac{1-\alpha}{2T} \right) \right) \right), & \frac{1-\alpha}{2T} < |f| < \frac{1+\alpha}{2T} \\ 0, & \text{elsewhere} \end{cases}$$

Solution 1

- Bandwidth of raised cosine pulse: $B_T = \frac{1+\alpha}{2T_b}$
- Bandwidth of raised cosine pulse:
 $B_T = \frac{(1+\alpha)R_b}{2} = (1 + \alpha)28\text{KHz}.$
- Excess bandwidth $\frac{\alpha R_b}{2} = \alpha \times 28 \text{ KHz}.$
- $\alpha = 0$: $B_T = 28 \text{ KHz}$, Excess bandwidth= 0.
- $\alpha = 0.25$: $B_T = 35 \text{ KHz}$, Excess bandwidth= 7 KHz
- $\alpha = 0.5$: $B_T = 42 \text{ KHz}$, Excess bandwidth= 14 KHz
- $\alpha = 0.75$: $B_T = 49 \text{ KHz}$, Excess bandwidth= 21 KHz
- $\alpha = 1$: $B_T = 56 \text{ KHz}$, Excess bandwidth= 28 KHz

Nyquist Pulses

- Sinc pulse: $p(t) = \text{sinc}(\pi R_s t)$
- Raised cosine pulse: $p(t) = \frac{\cos(\pi \alpha R_s t)}{1 - (2\alpha R_s t)^2} \text{sinc}(\pi R_s t)$
- $B_T = \frac{1+\alpha}{2T_s}$

7.7-1 In a PAM scheme with $M = 16$:

- (a) Determine the minimum transmission bandwidth required to transmit data at a rate of 12,000 bits/sec with zero ISI.
- (b) Determine the transmission bandwidth if Nyquist criterion pulses with a roll-off factor $r = 0.2$ are used to transmit data.

CW Problem

- 7.7-4** An audio signal of bandwidth 10 kHz is sampled at a rate of 24 kHz, quantized into 256 levels and coded by means of M -ary PAM pulses satisfying Nyquist's criterion with a roll-off factor $r = 0.2$. A 30 kHz bandwidth is available to transmit the data. Determine the best value of M .

Nyquist Criterion for Zero Intersymbol Interference: Frequency Domain

- Time-domain

$$\bar{p}(t) = p(nT_s) = \delta(n)$$

- Nyquist criterion for zero ISI (Frequency-Domain):
- Impulse train $\bar{p}(t) = p(t)\delta_{T_s}(t)$
- Fourier series of $\delta_{T_s}(t) = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} e^{jn2\pi f_s t}$
- Impulse train $\bar{p}(t) = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} p(t)e^{jn2\pi f_s t}$
- Fourier transform
- $\frac{1}{T_s} \sum_{n=-\infty}^{\infty} P(f - \frac{n}{T_s}) = 1$
- $\sum_{n=-\infty}^{\infty} P(f - \frac{n}{T_s}) = T_s$
- Or, $\sum_{n=-\infty}^{\infty} P(f + \frac{n}{T_s}) = T_s$
- Sum of spectra formed by repeating $P(f)$ spaced $1/T_s$ apart must be a constant, T_s .

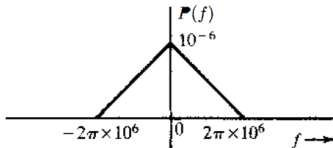
Nyquist Pulses

Is this true: All band-limited signals are Nyquist pulses?

7.3-4 The Fourier transform $P(f)$ of the basic pulse $p(t)$ used in a certain binary communication system is shown in Fig. P7.3-4.

- (a) From the shape of $P(f)$, explain at what pulse rate this pulse would satisfy Nyquist's criterion.
- (b) Find $p(t)$ and verify that this pulse does (or does not) satisfy the Nyquist's criterion.
- (c) If the pulse does satisfy the Nyquist criterion, what is the transmission rate (in bits per second) and what is the roll-off factor?

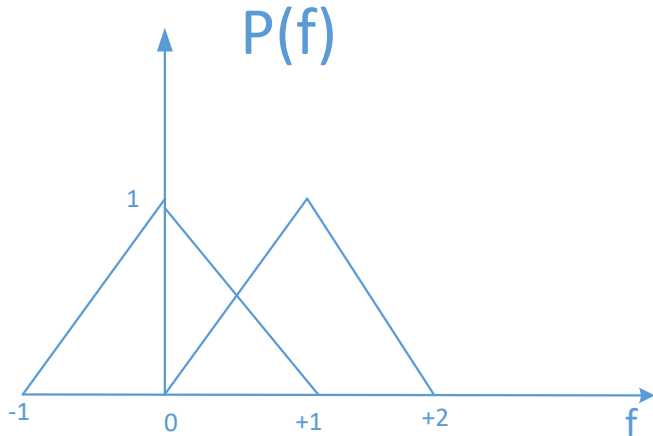
Figure P.7.3-4



CW Problem

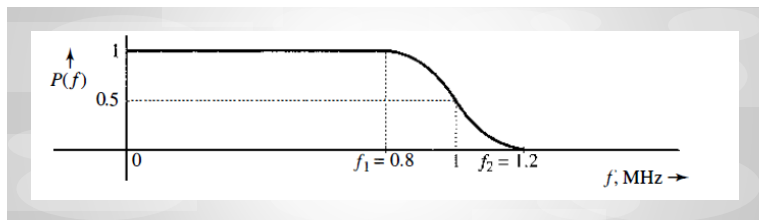
A message signal $m(t) = \sum p(t - kT)$, where $p(t) = \text{sinc}^2(\pi t)$ is used in binary communications and transmitted with pulse rate satisfying Nyquist criterion for zero-ISI. Sketch the frequency response of the pulse for $k = 0$ and $k = 1$ in the same plot.

Solution



CW Problem

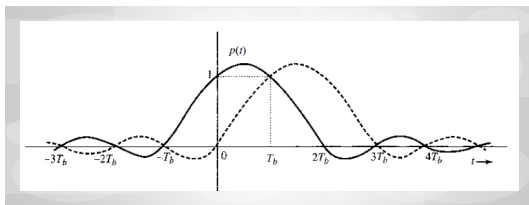
A pulse $p(t)$ whose spectrum $P(f)$ is shown in Fig. 1 satisfies Nyquist's criterion. If $f_1 = 0.8$ MHz and $f_2 = 1.2$ MHz, determine the maximum rate at which binary data can be transmitted by this pulse using Nyquist's criterion. What is the roll-off factor?



Nyquist Pulse with Controlled ISI: Second Criterion

- BW requirement is higher than minimum $R_b/2$.
- Widening the time duration: decreases the BW but increases the ISI
- Can we widen the duration of pulse in a controlled way such that BW is required and ISI can be compensated.

$$p(nT_b) = \begin{cases} 1, n = 0, 1 \\ 0, \text{for all other } n \end{cases}$$



Nyquist Pulse with Controlled ISI: Second Criterion

- Duobinary pulse: $p(t) = \frac{\sin(\pi R_b t)}{\pi R_b t(1 - R_b t)}$
- $P(f) = \frac{2}{R_b} \cos(\frac{\pi f}{R_b}) \Pi(\frac{f}{R_b}) e^{-j\pi f/R_b}$

Nyquist Pulse with Controlled ISI: Second Criterion

- At $t = 0$ one pulse is transmitted which affects the pulse at the succeeding pulse $t = T_s$.
- ISI at $t = T_s$ will be from the previous pulse transmitted at $t = 0$.
- How we can remove the ISI at $t = T_s$ from previous pulse.
- Let polar signaling is used.
- If sample value is positive: Present bit is positive and previous bit was also positive
- If sample value is negative: Present bit is negative and previous bit was also negative.
- If sample value is zero: the present bit is opposite of the previous bit.

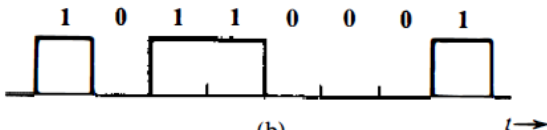
Bandpass Digital Modulation: ASK

- $m(t) = \sum a_k p(t - kT_b)$
- $p(t)$ is a rectangular pulse.
- If $p(t)$ is On-Off, i.e. $a_k = 0, 1$.
- $x(t) = m(t) \cos(2\pi f_c t)$
- Carrier amplitude is varied in proportion to $m(t)$

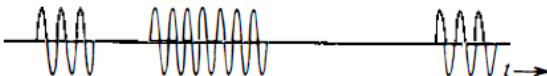
Amplitude Shift Keying



(a)



(b)



(c)

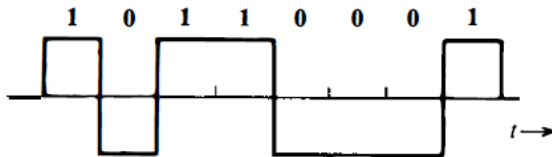
Bandpass Digital Modulation: PSK

- $m(t) = \sum a_k p(t - kT_s)$
- $p(t)$ is a rectangular pulse.
- If $p(t)$ is polar, i.e. $a_k = \pm 1$.
- $x(t) = m(t) \cos(2\pi f_c t)$
- Or $x(t) = m(t) \cos(2\pi f_c t)$ for digit 1
- $x(t) = -m(t) \cos(2\pi f_c t) = m(t) \cos(2\pi f_c t + \pi)$ for digit 0
- If the carrier phase is varied in proportion to $m(t)$

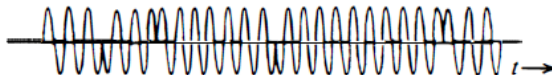
Bandpass Digital Modulation: FSK

- $m(t) = \sum a_k p(t - kT_s)$
- $p(t)$ is a rectangular pulse.
- $x(t) = m(t) \cos(2\pi f_{c1}t)$ for digit 1
- $x(t) = m(t) \cos(2\pi f_{c0}t)$ for digit 0
- If the carrier frequency is varied in proportion to $m(t)$

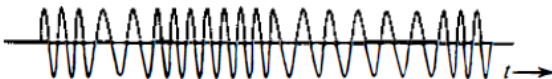
PSK (b) and FSK Waveform (c)



(a)



(b)



(c)

PSD: General approach

- $S_y(f) = |P(f)|^2 S_x(f)$.
- $S_x(f) = \frac{1}{T_b} [R_0 + 2 \sum_{n=1}^{\infty} R_n \cos n2\pi f T_b]$
- where $R_n = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_k a_k a_{k+n}$ is the auto-correlation of line encoded waveform.
- For proof: See section 8.2.1 B.P. Lathi 4th SE edition page 511 or section 7.2.1. B.P. Lathi 4th International edition.

Auto-Correlation of Polar line codes

Possible Values of $a_k a_{k+1}$

		a_k	
		-1	+1
a_{k+1}	-1	1	-1
	+1	-1	1

Auto-Correlation of Polar Line Codes

- What is R_n ?
- $R_n = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_k a_k a_{k+n}$
- $R_0 = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_k a_k^2$
- In polar signal 1 is transmitted as $+p(t)$ and 0 as $-p(t)$.
- $R_0 = \lim_{N \rightarrow \infty} \frac{1}{N} (N) = 1$
- $R_1 = \lim_{N \rightarrow \infty} \frac{1}{N} \left[\frac{N}{2} (1) + \frac{N}{2} (-1) \right] = 0$
- Thus, $R_n = 0, n \geq 1$

Auto-Correlation of On-Off codes

- $R_0 = \lim_{N \rightarrow \infty} \frac{1}{N} [\frac{N}{2}(1) + \frac{N}{2}(0)] = 1/2$
- For R_n :
- Product $a_k a_{k+n}$: $1 \times 1, 1 \times 0, 0 \times 1, 0 \times 0$
- $R_1 = \lim_{N \rightarrow \infty} \frac{1}{N} [\frac{N}{4}(1) + \frac{3N}{4}(0)] = 1/4$
- Thus, $R_n = 1/4, n \geq 1$

Auto-Correlation of Bipolar

- What is R_n ?
- $R_n = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_k a_k a_{k+n}$
- $R_0 = \frac{1}{2}$: show this
- $R_1 = -\frac{1}{4}$: show this
- $R_n = 0, n \geq 2$.

PSD of Line Encoding

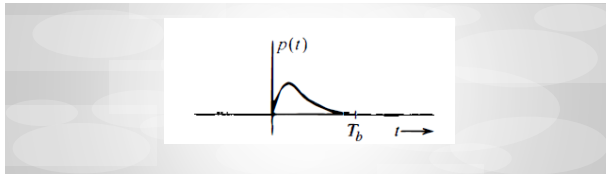


Figure: Pulse

- Baseband transmission: $y(t) = \sum a_k p(t - kT_s)$, where pulses are T_s unit of time apart.

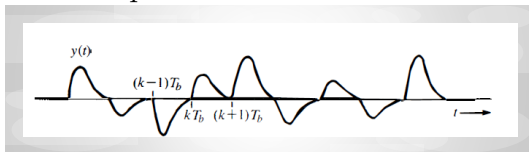


Figure: Pulse

- What is PSD of $y(t)$?

PSD: General approach

- Since a_k depends on line encoding, it is easier to find PSD of $x(t)$ which is an impulse train with impulses existing at T_s

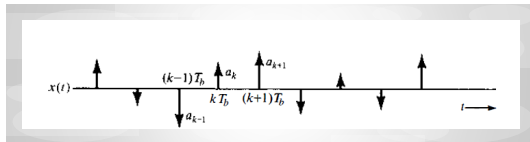


Figure: Pulse

- $S_y(f) = |P(f)|^2 S_x(f)$.
- $S_x(f) = \frac{1}{T_b} [R_0 + 2 \sum_{n=1}^{\infty} R_n \cos n2\pi f T_b]$
- where $R_n = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_k a_k a_{k+n}$ is the auto-correlation of line encoded waveform.
- For proof: See section 8.2.1 B.P. Lathi 4th SE edition page 511 or section 7.2.1. B.P. Lathi 4th International edition.